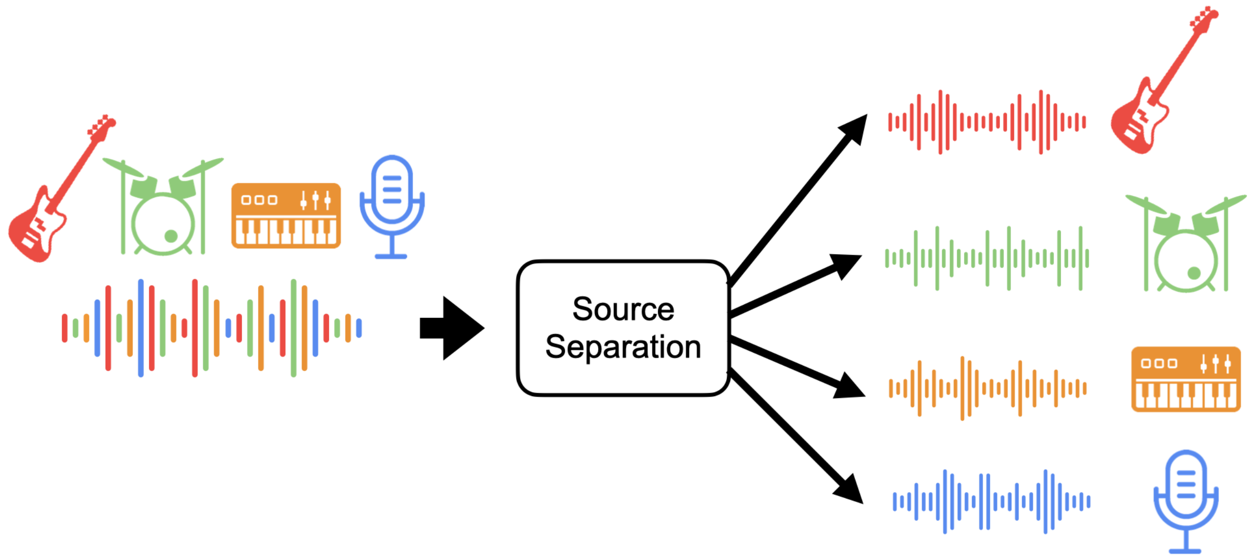
Audio Source Separation

Independent Research Paper, Fall 2023

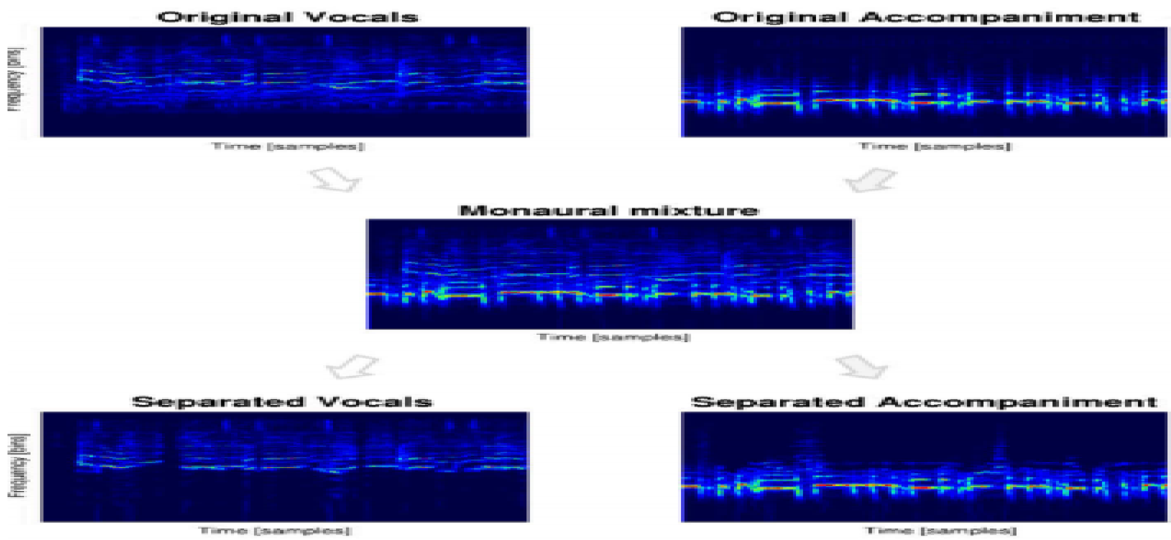
Description: In 1953, British Scientist Colin Cherry identified a phenomenon that we do naturally as human beings. Picture yourself in an environment where you're surrounded by multiple stimuli in a given area but you are still capable of focusing on one specific stimulus. This is referred to as the Cocktail Party Problem where we are capable of blocking other stimuli (usually auditory), in order to focus on an individual one. Throughout recent years, Engineers have figured out how to emulate the Cocktail Party Problem using computer software and hardware through certain algorithms. Whether it be music, bio acoustics, or human voice separations, There are many uses for audio source separation to be incorporated for essentials usage or job oriented tasks

Why Source Separation?

In the book "**Audio Source Separation and Speech Enhancement**" in the context of speech, music, and ambient audio processing, there are several application issues that give rise to the problems of source separation like voice signals being affected by background noise, reverberation, and/or intervening speakers. Aside from degrading voice quality, these occurrences may also affect speech intelligibility and automated speech recognition (ASR) performance. In these kinds of situations, source separation and speech improvement are necessary to separate or amplify the near-end speaker's voice from other speakers during conversation over cell phones or hands-free systems. Strong distant-microphone ASR requires source separation and speech augmentation, two additional preprocessing stages that are included in today's personal assistants, automobile navigation systems, televisions, video game consoles, and medical dictation devices(**Vincent, E., Virtanen, T., & Gannot, S. (2018) pg.3**).

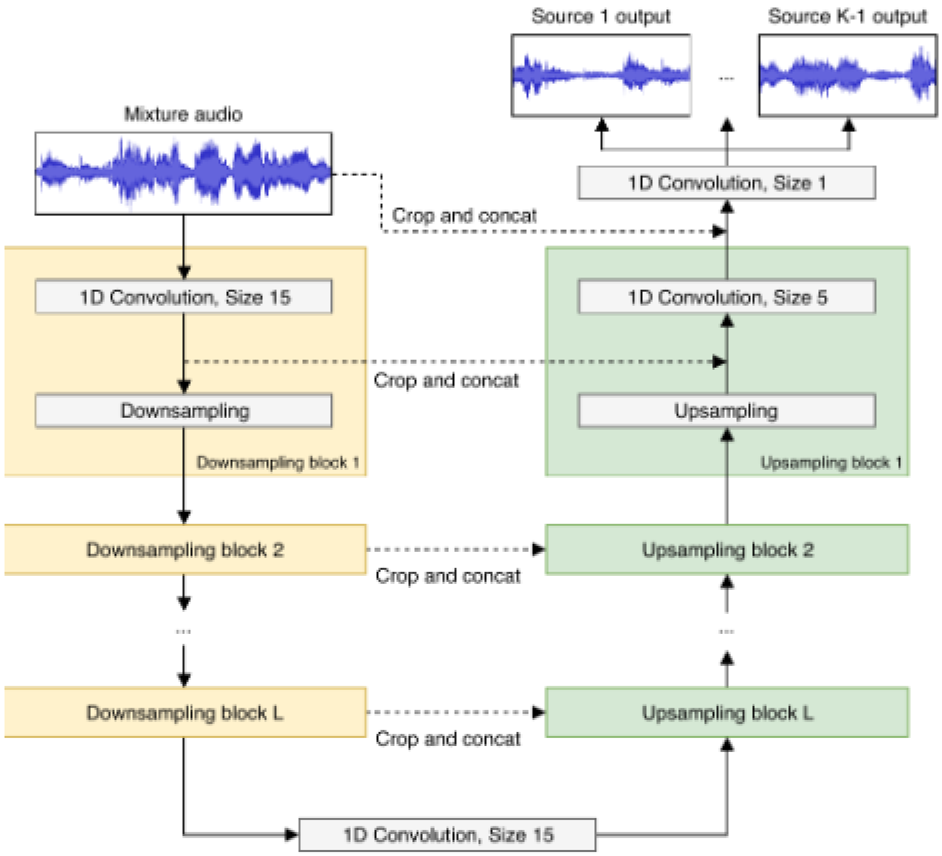


Deep Neural Networks: Neural networks, a type of artificial intelligence, teach computers to process information similarly to the human brain. Deep learning is a type of machine learning approach that mimics the structure of the human brain by using networked nodes or neurons arranged in layers. A simple deep learning technique that provides a database of songs and individual voices and instruments while creating spectrograms of each audio mixture is how MIT student Andrew Simpson solves the well-known cocktail party problem involving music (**Emerging Technology from the arXiv., 2015**).



A spectrogram is a graphical depiction of a signal's strength, or "loudness," at different frequencies within a given waveform across time. The model then simply takes the input of the given spectrogram mixture, then outputs the desired output whether being the voice or music itself (**Emerging Technology from the arXiv., 2015**).

Unet: The U-net is a Convolutional Neural Network (CNN) architecture that utilizes skip connections as an encoder/decoder. For each stem, a U-net is used to estimate a soft mask stated in the article **written by Romain Hennequin , Anis Khlif , Felix Voituret , and Manuel Moussallam[].**



Algorithms:

MDX23(Music Demixing Track 23)

From the github repository by ZFTurbo, this model is based on Ultimate Vocal Remover project Demucs4, and MDX neural net structures.

Demucs HT 4

In the article "**Hybrid Transformers for Music Source Separation**", Its talks about this model using a convolutional neural network architecture called Wave-u-net and features a hybrid/ spectrogram/waveform separation model using a Cross-Domain Transformer encoder(**Rouard, S., Massa, F., & Défossez, A. (2023)**).

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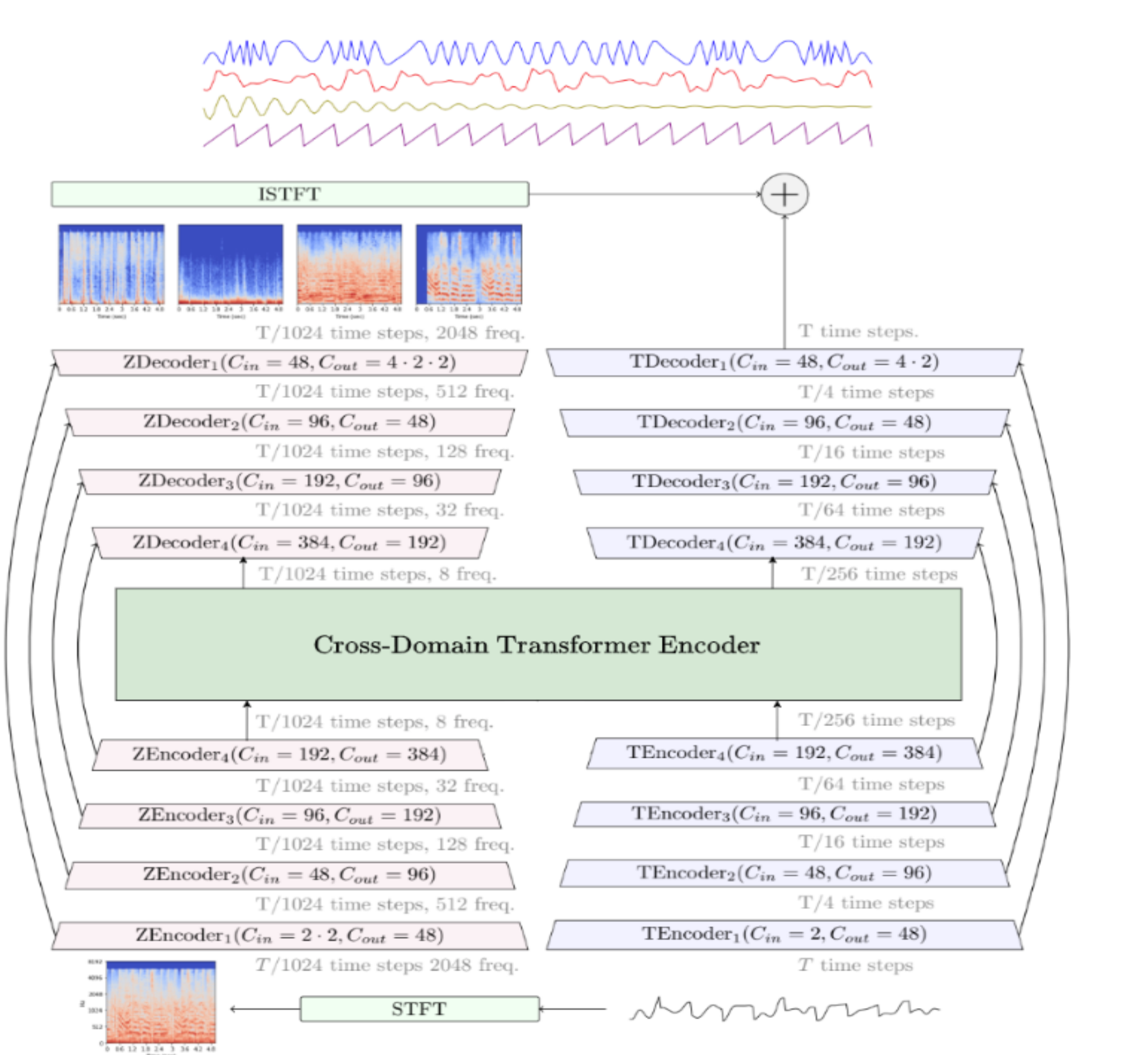

Transformers are known for their ability to process sequential data, such as audio or language. The encoder breaks down the input audio signal into smaller, hierarchically structured representations within the context of the transformer architecture. The transformer encoder is said to be "cross-domain" if it has been trained to recognize and extract features from different kinds of audio signals without requiring specific training for each kind. Other parts of the Demucs HT 4 model use the features extracted from the audio stream by the transformer encoder to identify individual sound sources, including bass, drums, vocals, and other elements in the mixed audio such as guitars and piano.

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Demucs 3

Its ability to split mixed audio into three different stems or sources is discussed by Alexandre Fossez.Given that they are adept at processing sequential data, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) are most likely the neural network architectures that will form the basis of it (Rouard, S., Massa, F., & Défossez, A. (2023)).

Spleeter

one of the earliest source separation models that the public could access and use. Splitter uses a deep learning architecture called TensorFlow stated in the article to be among the many deep learning libraries available, the most well-known was initially developed by Google researchers.

Quality Comparison

MultiSong Dataset

Algorithm	SDR bass	SDR drums	SDR other	SDR vocals	SDR instrumental
MVSEP MDX23	12.5034	11.6870	6.5378	9.5138	15.8213
Demucs HT 4	12.1006	11.3037	5.7728	8.3555	13.9902
Demucs 3	10.6947	10.2744	5.3580	8.1335	14.4409
MDX B	---	----	---	8.5118	14.8192

- Note: SDR - signal to distortion ratio. Larger is better.

MusDB Dataset

Method	All	Drums	Bass	Other	Vocals
Hybrid Demucs	**7.33**	**8.04**	**8.12**	5.19	7.97
KUIELAB-MDX-Net	7.24	7.17	7.23	**5.63**	**8.90**
Music_AI	6.88	7.37	7.27	5.09	7.79

Overall, Demucs would be preferred for drum and bass music samples, but the MDX algorithm would perform better when it came to vocals and accompaniment.

Cell Phone Noise Suppression: Cell phone noise suppression, which reduces unwanted background noises during calls to improve call quality, is a crucial feature of mobile devices. Cell phone calls can be disrupted by a variety of ambient noises, such as wind, traffic, crowd chatter, etc., which makes it hard for the parties to communicate clearly. Reducing these disturbances is the aim of noise suppression in order to improve the clarity of conversations. Three engineers developed an algorithm that performs blind source separation, which means that random noise is used as input rather than original samples taken from a database. Unwanted noise is then processed by doing the following:

Maximizing the entropy of the speech signal

$$H(y_i) = -\mathrm{E}[\log(f_{y_i}(y_i))],$$ * the negative entropy multiplied by the PDF(Probability Density Function)

PDF of the output

$$f_{y_i}(y_i) = f_{x_i}(x_i)/\det(J),$$

where J is the Jacobian of the unmixing network. Maximizing the entropy leads to maximizing $\mathrm{E}[\log(\det(J))]$. A stochastic gradient rule can be derived from this to obtain the unmixing filter updates as follows

$$\Delta w_{ij}(k) \propto \hat{y_i}\, u_j(n-k), \tag{3}$$

where

$$\hat{y_i} = \frac{\partial}{\partial y_i}\left(\frac{\partial y_i}{\partial u_i}\right). \tag{4}$$

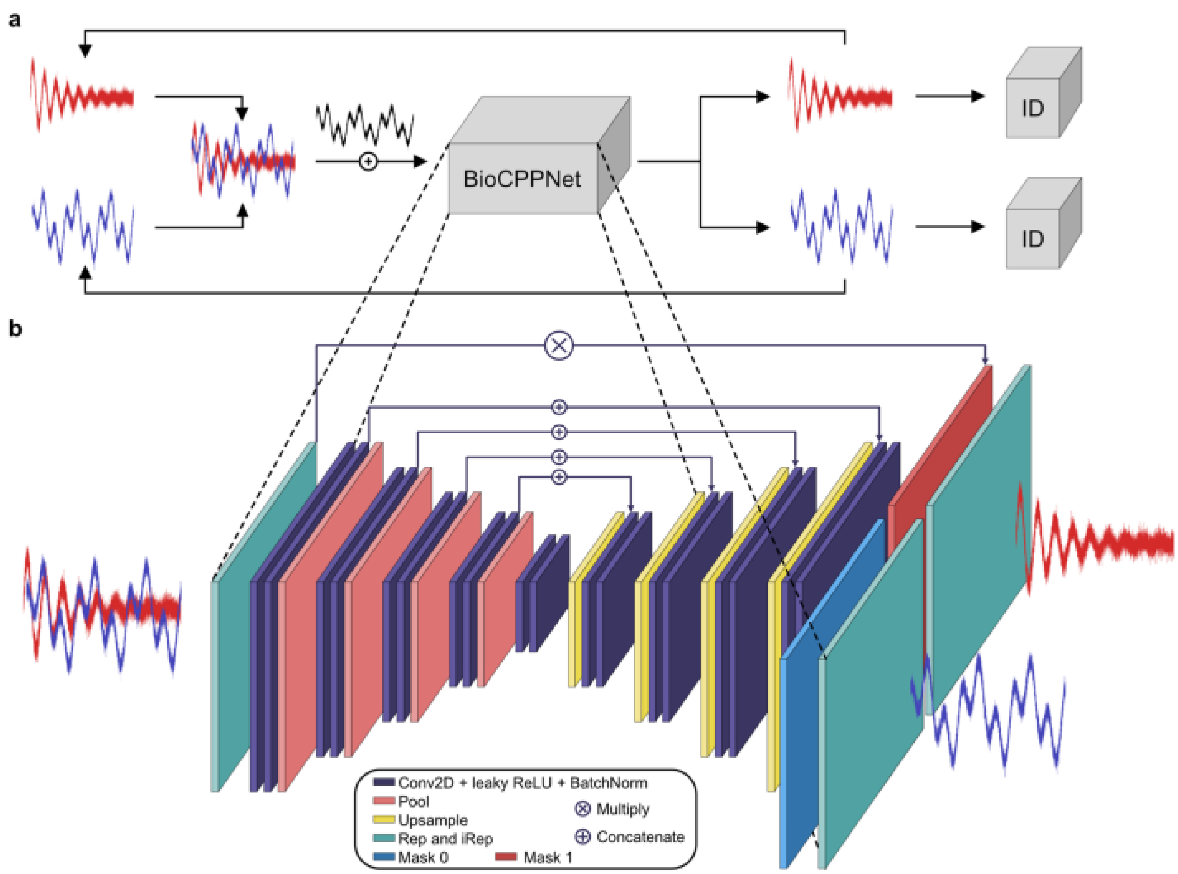
The separated sources u_i are obtained by applying these unmixing filters w_{ij} to the mixtures x_j

$$u_i(n) = x_i(n) + \sum_{k=0}^{P-1} w_{ij}(k) x_j(n-k). \tag{5}$$

where P is the length of the unmixing filters.

(Rouard, S., Massa, F., & Défossez, A. (2023), pg2)

BioCPPNet: Bioacoustics is the study of sound in biological organisms, including insects, animals, and their environments. For instance, in a forest, you might hear a mix of insect sounds, bird calls, and rustling leaves. The goal is to precisely distinguish and identify each sound, which can be done with the help of an algorithm known as BioCPPNet. BioCPPNet is a comprehensive, unique pipeline for permutation-invariant bioacoustic source separation across species. An article discussing the algorithm's inner workings states that it is applied to specific bioacoustics datasets, such as fruit bat cries, dolphin whistles, and macaque coos' vocalizations (Bermant, P. C., 2021).



(a) Schematic Overview of BioCppNet PipeLine

(b) BioCppNet Architecture

Conclusion: The use of machine learning, deep neural networks, and advanced signal processing algorithms is contributing to the progress of technology by improving the precision, resilience, and immediate application of audio source separation techniques. As audio source separation develops, more innovations are anticipated, expanding its applications beyond sectors and academic fields and ultimately improving our experience with sound and audio data.

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